

Home Lab Project Documentation week 4 day 1

Task: Perform a basic network mapping scan from Kali Linux to ensure the Ubuntu target is visible. Take snapshots of all configured virtual machines to establish a recovery baseline.

Part 1 — Basic network mapping

From your **Kali Linux VM**, verify that the **Ubuntu protected system** is reachable on your isolated network.

Based on your current lab configuration:

System	IP Address	Role
Kali Linux	192.168.64.2	Testing/Analyst system
Ubuntu	192.168.64.3	Protected/Target system
Network	192.168.64.0 /24	Isolated virtual network

A basic Nmap scan would be:

```
nmap -sn 192.168.64.0/24
```

```

ubuntu@ubuntu:~$ ifconfig
enp0s1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.64.3 netmask 255.255.255.0 broadcast 192.168.64.255
    inet6 fe80::1863:23ff:fe39:a42c prefixlen 64 scopeid 0x20<link>
    inet6 fda6:6bc3:cab9:adbc:1863:23ff:fe39:a42c prefixlen 64 scopeid 0x0
<global>
    ether 1a:63:23:39:a4:2c txqueuelen 1000 (Ethernet)
    RX packets 76 bytes 41132 (41.1 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 132 bytes 18156 (18.1 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 132 bytes 11374 (11.3 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 132 bytes 11374 (11.3 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

```

You should expect to see the Ubuntu system at:

192.168.64.3

Actually when I nmap -sn 192.168.64.0, ubuntu is not visible, but when I ping 192.168.64.3 or use a direct check: nmap -sn 192.168.64.3, I get a response.

The reason for not getting a response when using the command nmap -sn 192.168.64.3 because the network type was Nat.

```

(kali㉿kali)-[~]
└─$ nmap -sn 192.168.64.0
Starting Nmap 7.94 ( https://nmap.org ) at 2026-09-14 10:01 PDT
Note: Host seems down. If it is really up, but blocking our ping probes, try
-Pn
Nmap done: 1 IP address (0 hosts up) scanned in 3.01 seconds

```

```
(kali㉿kali)-[~]  
$ ping -c 4 192.168.64.3  
PING 192.168.64.3 (192.168.64.3) 56(84) bytes of data.  
64 bytes from 192.168.64.3: icmp_seq=1 ttl=64 time=0.713 ms  
64 bytes from 192.168.64.3: icmp_seq=2 ttl=64 time=1.01 ms  
64 bytes from 192.168.64.3: icmp_seq=3 ttl=64 time=0.941 ms  
64 bytes from 192.168.64.3: icmp_seq=4 ttl=64 time=0.904 ms  
  
— 192.168.64.3 ping statistics —  
4 packets transmitted, 4 received, 0% packet loss, time 3016ms  
rtt min/avg/max/mdev = 0.713/0.892/1.012/0.110 ms
```

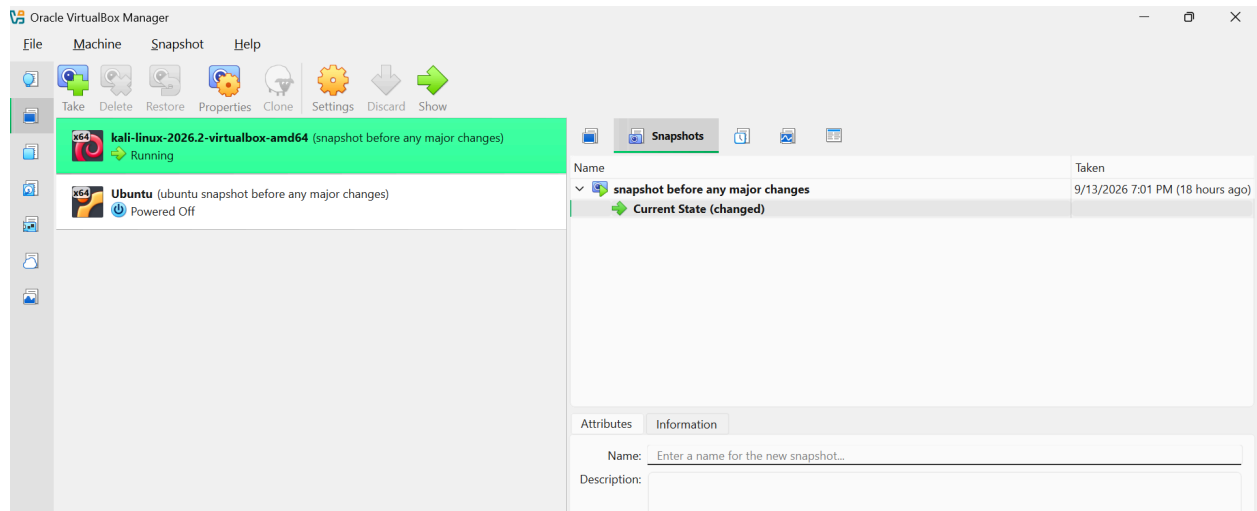
```
(kali㉿kali)-[~]  
$ nmap 192.168.64.3  
Starting Nmap 7.94 ( https://nmap.org ) at 2026-09-14 09:47 PDT  
Nmap scan report for 192.168.64.3  
Host is up (0.00045s latency).  
Not shown: 999 closed tcp ports (conn-refused)  
PORT      STATE SERVICE  
22/tcp    open  ssh  
  
Nmap done: 1 IP address (1 host up) scanned in 0.06 seconds
```

This confirms that Kali can see the Ubuntu target and establishes the starting point for later security testing.

Part 2 — Create VM snapshots

After confirming the network is working, shut down or pause the VMs as appropriate and create snapshots of your configured virtual machines.

Snapshots in Windows 11



UTM does not support taking snapshot, however there is a different way of saving the current state of the operating system.

For example:

- **Kali Linux — Week 4 Baseline**
- **Ubuntu — Week 4 Baseline**

The snapshots give you a known-good recovery point before you begin packet capture, reconnaissance, vulnerability assessment, and later testing.

What is being documented

Day 1 documentation includes:

1. Screenshot of Kali's network configuration.
2. Screenshot of `nmap -sn 192.168.64.0/24`.
3. Screenshot showing Ubuntu at `192.168.64.3`.
4. Screenshot of the Kali VM snapshot.
5. Screenshot of the Ubuntu VM snapshot.
6. Short statement confirming that the baseline was successfully established.

Important: This task is primarily a **baseline verification**, not a vulnerability scan. I am simply proving that the intended systems can see each other before moving into the Week 4 Wireshark and traffic-analysis work.